

Software Requirements Specification (SRS)

Field	Value
Project Name	Astro Code
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Class	10CTC
Date	Fri Jul 3, 2026
Version	1.0
GitHub Repository	https://github.com/astrogames06/astro-code-app

1. Introduction

1.1 Purpose

This document is the software requirements specification for the Astro Code app. It outlines the functional and non-functional requirements and the system design.

The purpose of this document is to clearly define what the system will do, how it will behave and how the user interacts with it. It is intended to developers and testers involved in the design of the system.

This SRS describes the requirements for the application so that it can be implemented, tested, and maintained in a structured and consistent way.

1.2 Product Overview

Astro Code is an app that teaches people to create games in Python. The app is made for students and teachers with an in built classroom system to make teaching and learning an ease. It solves the problem for students since it allows them to have a simple application to build, run, test and share games, and for teachers who are then able to assign work and set up classrooms easily.

1.3 Scope

In Scope

- User login and signup
- Classroom teacher/student system
- AI chat helper
- Saving user data to database

Out of Scope

- Online chat
 - Live sharing of code
 - Payment systems
 - Mobile app version
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1.4 Definitions, Acronyms and Glossary

Term	Definition
SRS	Software Requirements Specification
MVP	Minimum Viable Product
CRUD	Create, Read, Update, Delete
Database	Organised storage of application data
Primary Key (PK)	Unique identifier for a record
Foreign Key (FK)	Links one table to another

1.5 Technology Stack and Platform

- Frontend: React
 - Backend: Bun
 - Python Execution: Pyodide
 - Database: Supabase
 - Hosting: Vercel
 - Version Control: GitHub
 - AI Assistance: Lovable, GitHub Copilot, Cursor, ChatGPT
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2. Overall Description

2.1 Product Perspective

Astro Code is a web-based programming platform. It depends on Supabase for storing user data and Vercel for web hosting. It also uses Pyodide to execute the python code.

2.2 User Characteristics

User Type 1

- Description: Teachers wanting to set up an online Python classroom for students.
- Technical Skill: Intermediate.
- Needs: The app provides a classroom system where the teacher can create classrooms for each class and year and assign custom tasks.

User Type 2

- Description: A student connecting to their teachers classroom.
 - Technical Skill: Beginner
 - Needs: The app provides a way for the student to connect to teachers classroom and start working on assigned tasks and share them with their teacher and fellow students.
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2.3 Operating Environment

- Browser: Chrome, Edge, Safari, Firefox
 - Devices: Laptop, Desktop
 - Hosting: Vercel
 - Database: Supabase
 - Internet Required: Yes
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2.4 Design and Implementation Constraints

- Time: Limited time to design and build the project.
 - Budget: No budget available, so only free tools and services were used.
 - Free tier limits: The system relies on free-tier services such as Supabase and Vercel, which impose usage and storage limits.
 - Other constraints: The application uses specific technologies such as React, Bun, Supabase, and Vercel, which restrict certain implementation options.
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2.5 Assumptions and Dependencies

- Users have a internet connection
 - Users have a modern browser
 - Users have basic computer skills
 - Supabase & Vercel services are still available
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3. Functional Requirements

3.1 Functional Requirements List

ID	Requirement
FR-1	The system shall allow users to create an account.
FR-2	The system shall allow users to sign in to their account.
FR-3	The system shall allow teachers to create a classroom.
FR-4	The system shall allow students to join the teachers classroom via a classroom code.
FR-5	The system shall allow teachers to create a new task.
FR-6	The system shall allow teachers to assign a custom task to a specific class.

ID	Requirement
FR-7	The system shall allow students to view their assigned tasks.
FR-8	The system shall allow students to open the task and complete it.
FR-9	The system shall allow teachers to view the students progress on the task.
FR-10	The system shall allow students to share their project with fellow students.

3.2 User Stories

- As a student, I want to join a classroom so that I can access my assigned tasks.
- As a student, I want to view my assigned tasks so that I am able to complete them.
- As a student, I want to create a new game so that I am able to share it with my teacher and fellow students.
- As a teacher, I want to create a new classroom so that I am able to add all my students to it.
- As a teacher, I want to create a new task so that I can assign it to the students to complete.

3.3 Context / Use Case Summary

1. Teacher creates a new task and assigns it to the class.
2. Student views the assigned task and begins working on it.
3. Student completes the task and submits/shares it with the teacher.
4. Teacher views the student's progress on the task.
5. Teacher accesses the submitted project via a shared link and tests the game.

4. Non-Functional Requirements

Quality	Requirement
Performance	Pages load quickly, python code executes fast.
Usability & Accessibility	Simple interface for students and teachers.
Security & Privacy	Secure login and secure database.
Reliability	Data is saved consistently and retrieved correctly.
Maintainability	Clean code structure for easy updates.

5. System Design

5.1 Data Dictionary

Table: Teachers Classroom (classroom)

Field	Type	Key	Rules / Notes
id	uuid	PK	The unique identifier for each classroom.
teacher_id	uuid	FK	The unique identifier for the teacher that owns the classroom.
name	text		The name that is given to each classroom.
code	text		The code for each classroom that allows the students to join.

Table: Students Game Project (projects)

Field	Type	Key	Rules / Notes
id	uuid	PK	The unique identifier for each students game.
user_id	uuid	FK	The students unique identifier, who is the owner of the game.
name	text		The name given to the students game.
code	text		The Python code for the students project.
assets	jsonb		The images and assets for the students game.
is_public	bool		Determines if the students project is public meaning others can play and test it.

Table: classroom_assignments

Field	Type	Key	Rules / Notes
id	uuid	PK	The unique identifier for each classroom assignment.
classroom_id	uuid	FK	The classrooms unique identifier, which class the task is assigned to.
mission_slug	text		The name the database uses to identify the mission.

Table: classroom_memberships

Field	Type	Key	Rules / Notes
id	uuid	PK	The unique identifier for each classroom membership.
classroom_id	uuid	FK	The classrooms unique identifier.
student_id	uuid	FK	The students that are in the classroom, their unique identifier.

Table: classroom_student_progress

Field	Type	Key	Rules / Notes
id	uuid	PK	The unique identifier for each classrooms student progress.
classroom_id	uuid	FK	The classrooms unique identifier.

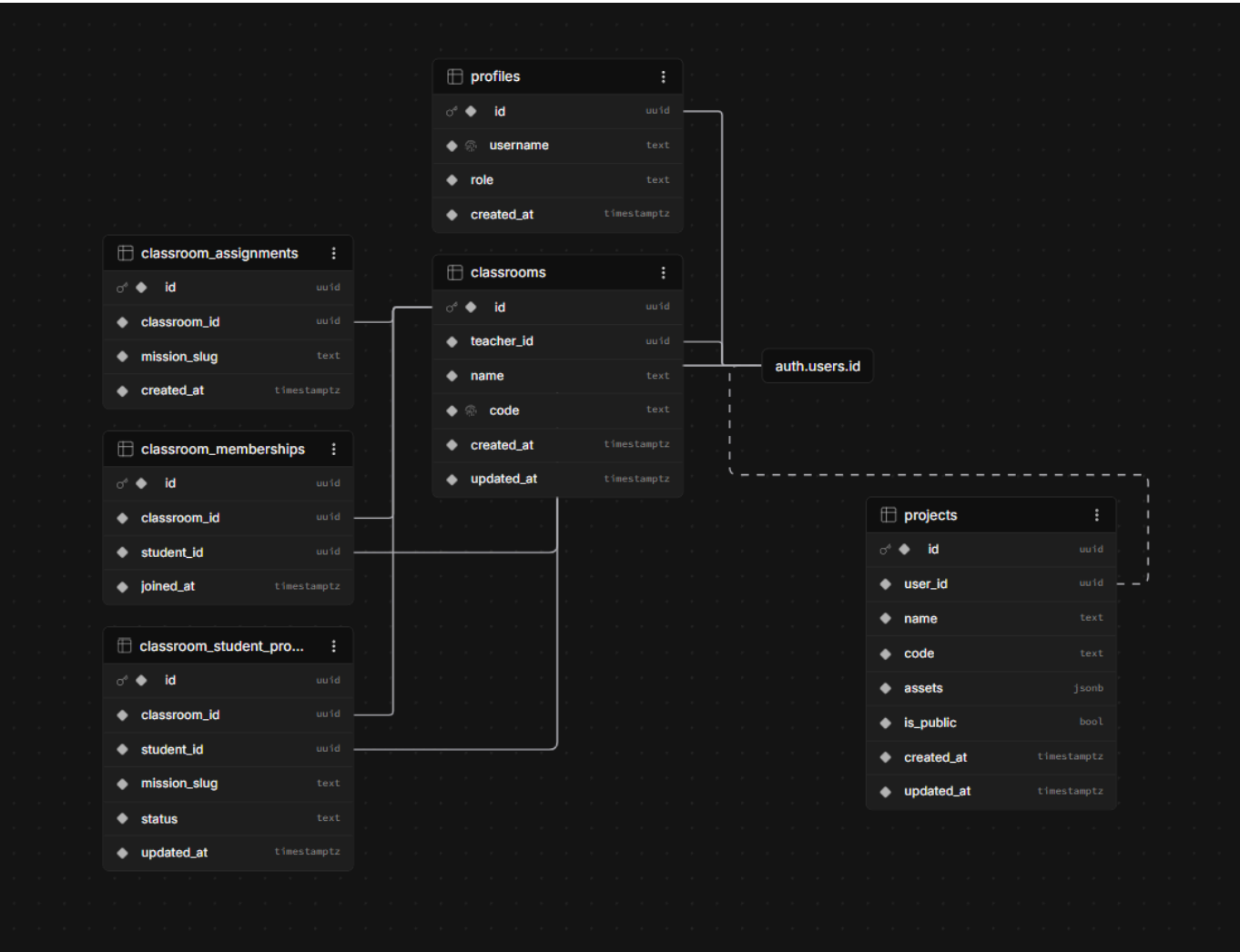
Field	Type	Key	Rules / Notes
student_id	uuid	FK	The students that are in the classroom, their unique identifier.
mission_slug	text		The name the database uses to identify the mission.
status	text		The students status on the assigned mission.

Table: profiles

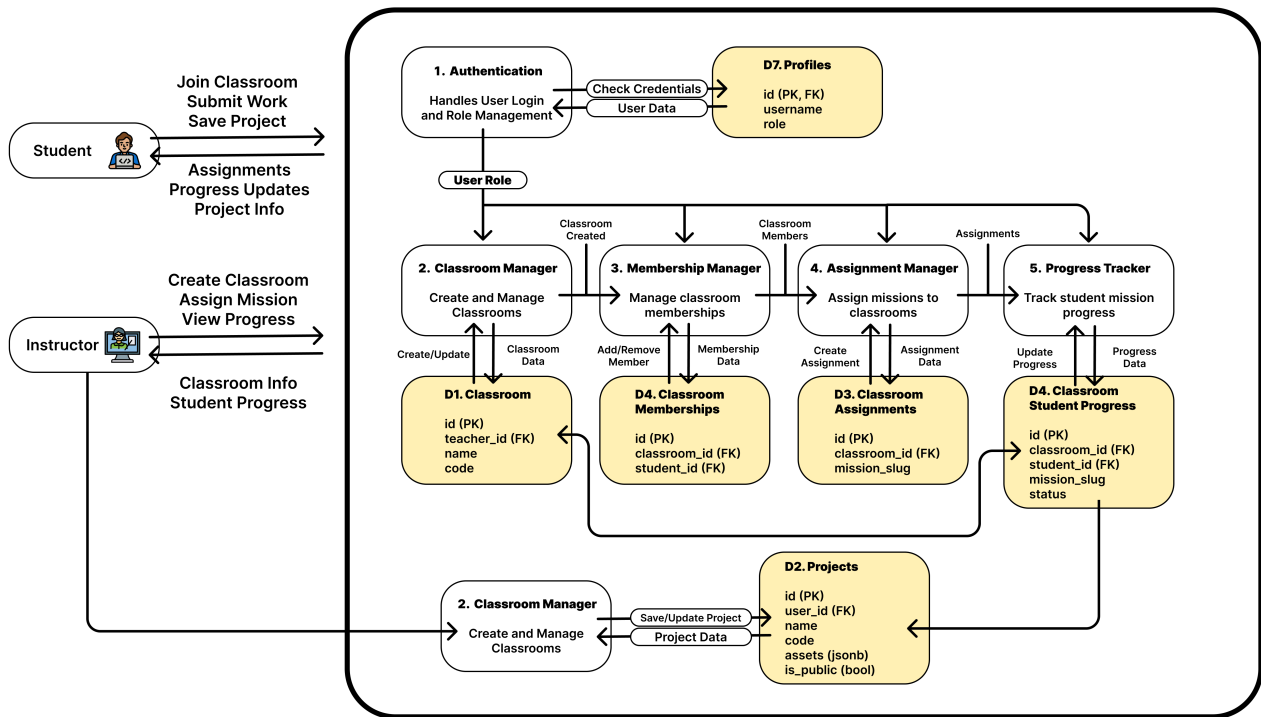
Field	Type	Key	Rules / Notes
id	uuid	PK FK	The unique identifier for the users profile.
username	text		The users unique username.
role	text		Wether the user is a student or teacher.

5.2 Entity Relationship Diagram (ERD)

Insert your ERD image.

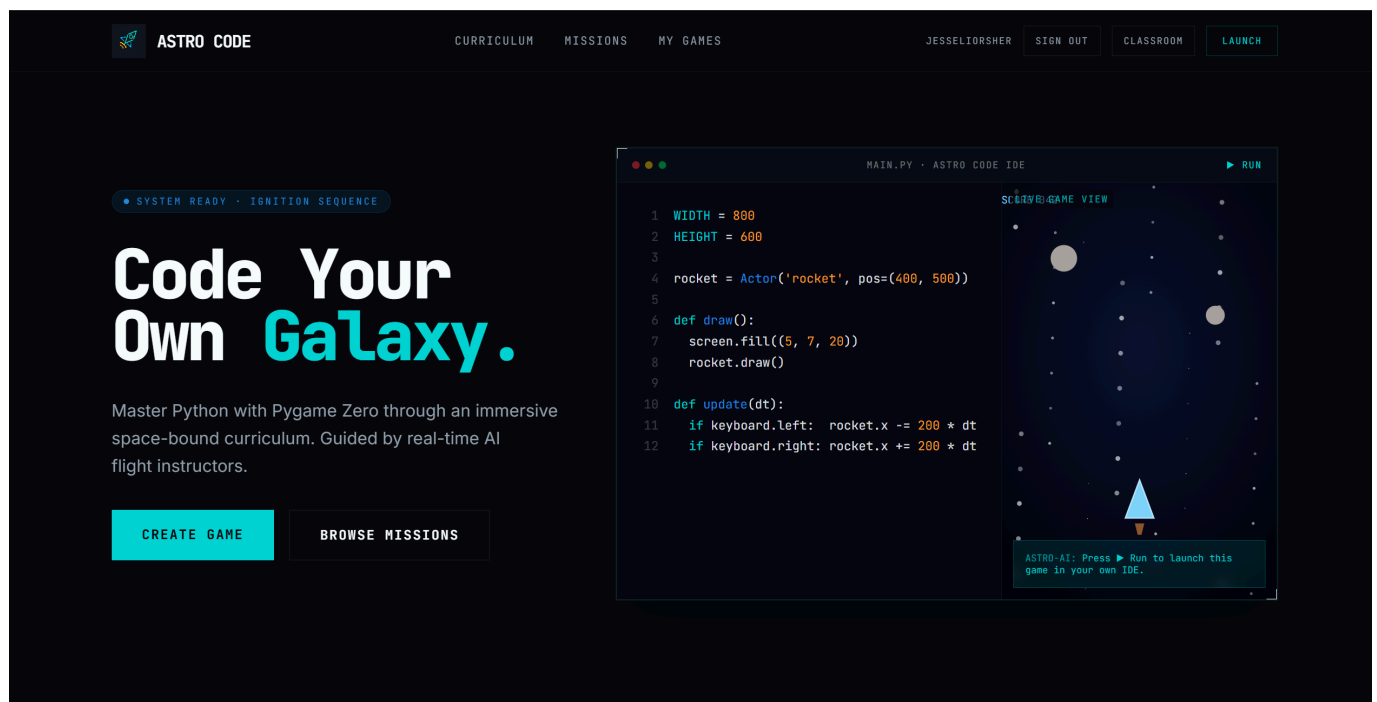


5.3 Data Flow Diagram (DFD)



5.5 User Interface Design

Home Page



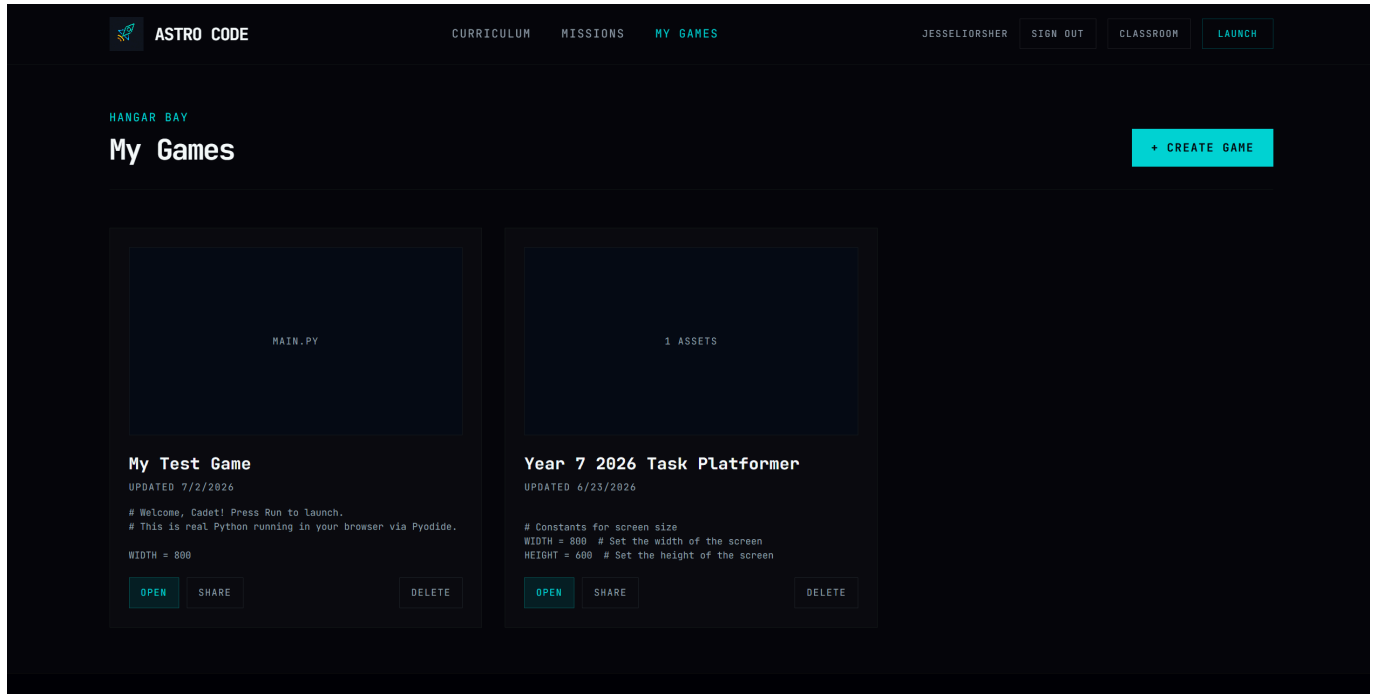
Purpose: Allows the user to access the Astro Code application and navigate to its main features.

Key Elements:

- Navigation Bar
 - Classroom Button
 - Launch Button

- Code Example
- Create Game Button
- Footer

Launch Page



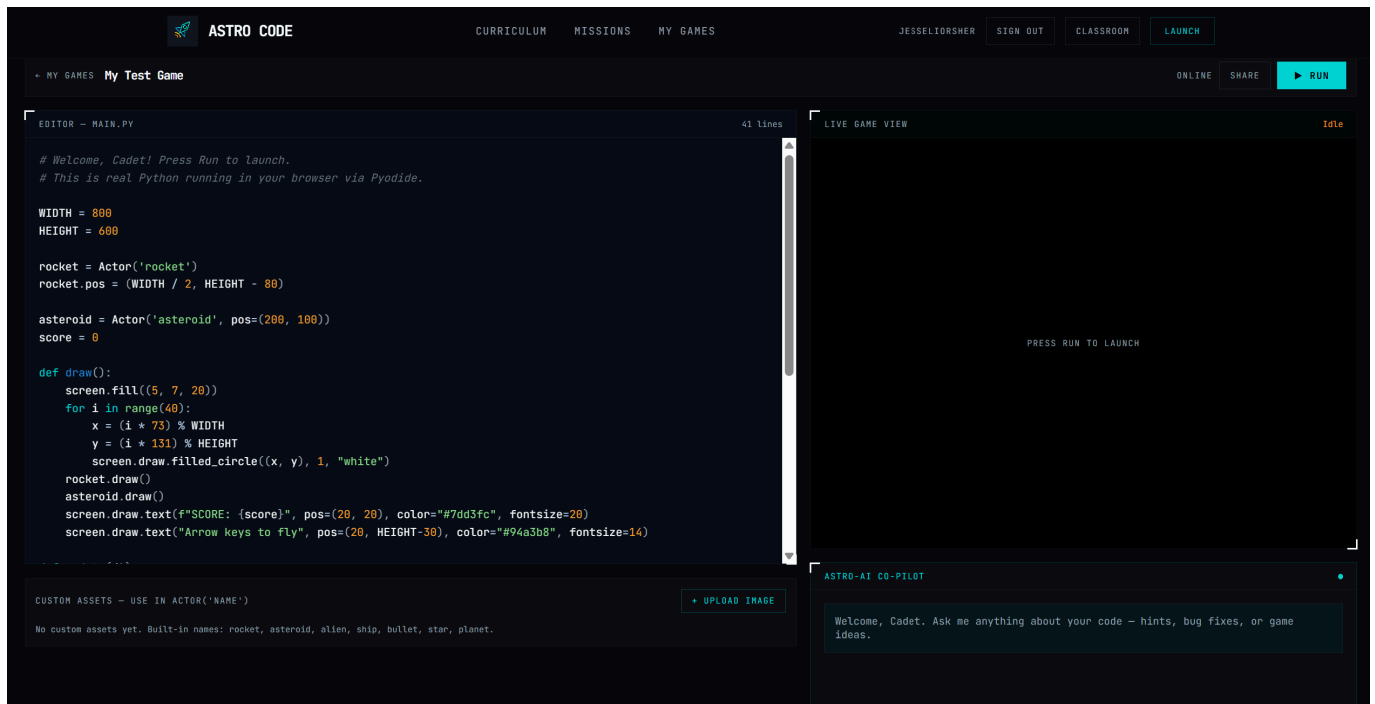
Purpose:

- To show all the users created games.

Key Elements:

- Navigation Bar
- Game Preview Container
- Create Game Button
- Open Game Button
- Delete Game Button
- Share Game Button
- Footer

Integrated Development Environment Page



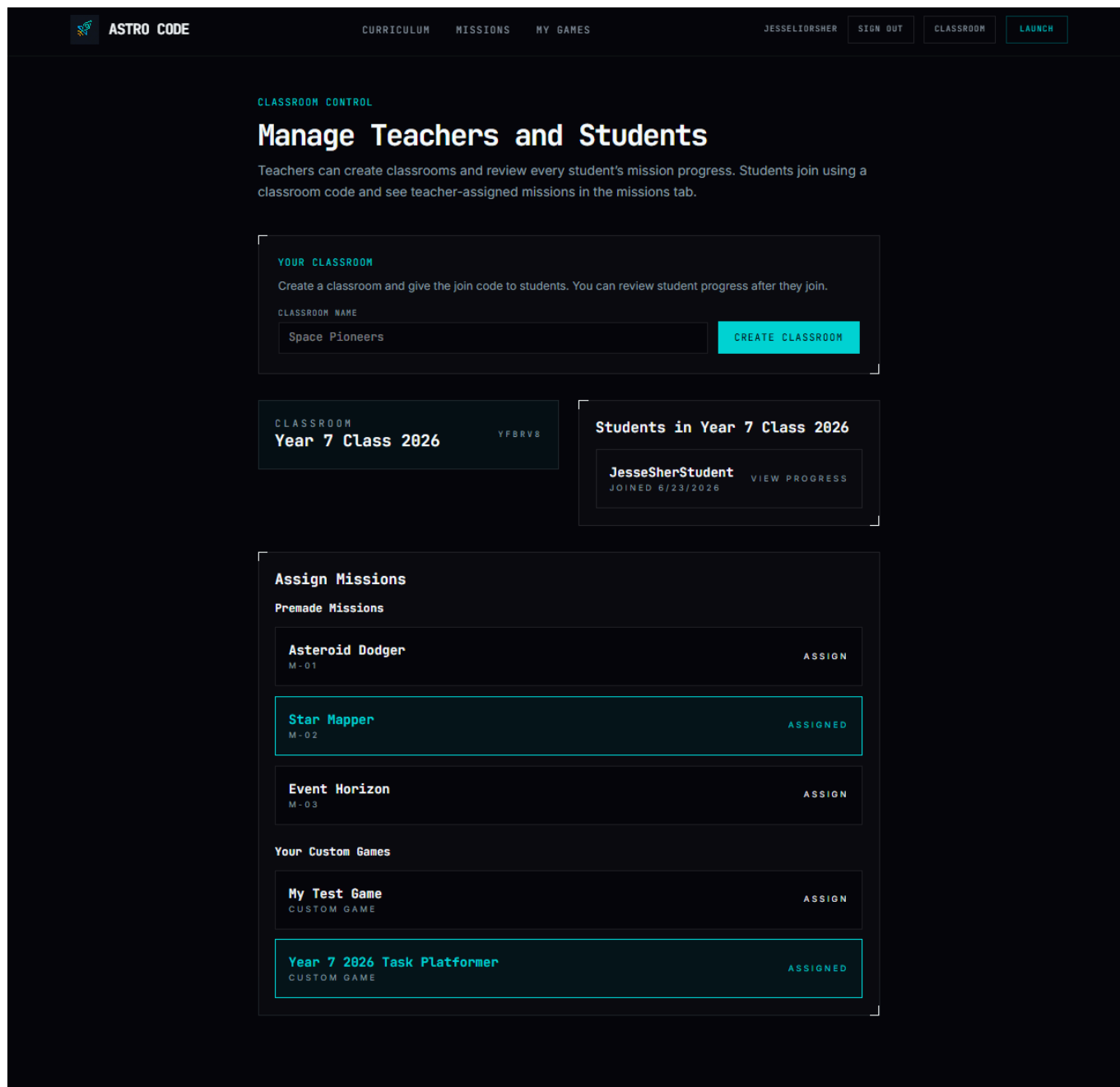
Purpose:

- For the user to be able to write code, upload assets and share their game.

Key Elements:

- Navigation Bar
- Code Editor Text Area
- Game Title
- Share Game Button
- Play Game Button
- Assets Uploader
 - Asset Upload Button
 - Preview Of Uploaded Assets
- Astro-AI chat
 - Input Textbox
 - Send Message Button
 - Chat Preview Box
- Footer

Classroom Page for Teacher



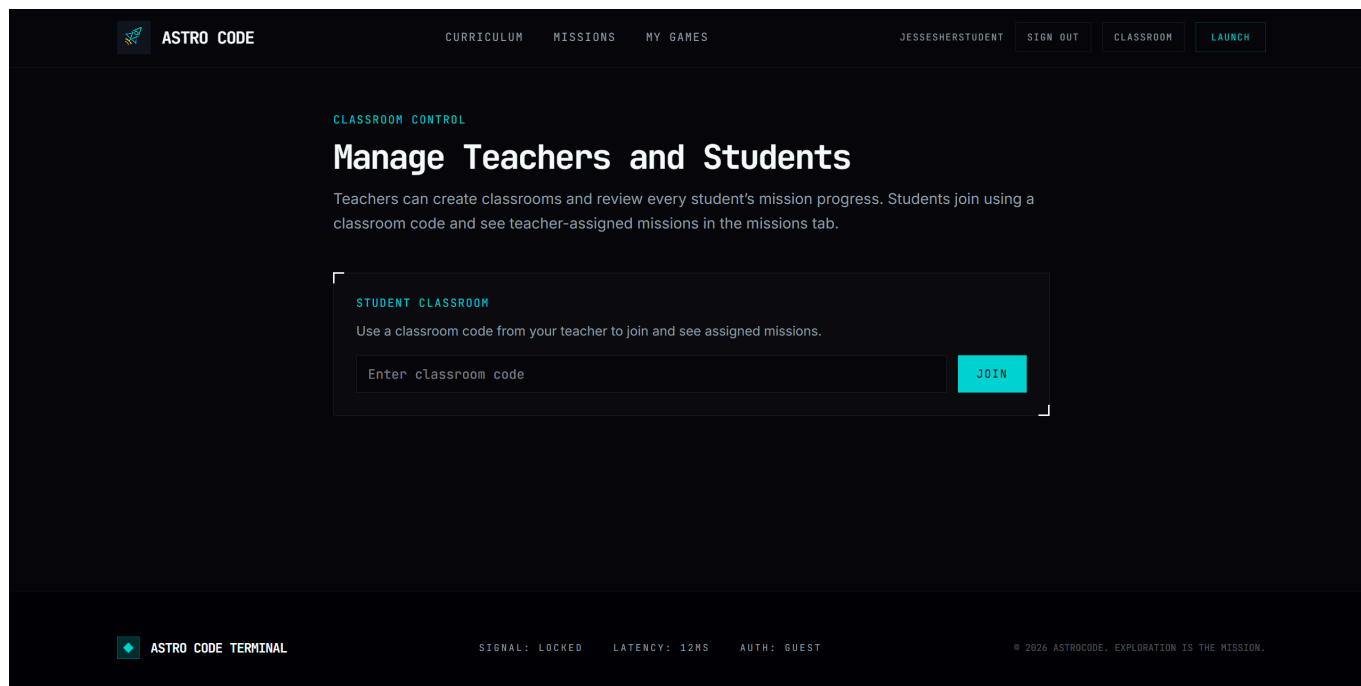
Purpose:

- For the teacher to be able to see classrooms, students and assign tasks.

Key Elements:

- Navigation Bar
- Create Classroom Button
- Classroom Name Input Box
- List of Classrooms
 - Code to Join the Classroom
 - List of Students in the Classroom
 - List of assigned mission for the Classroom
- Footer

Classroom Page for Student Before Joining



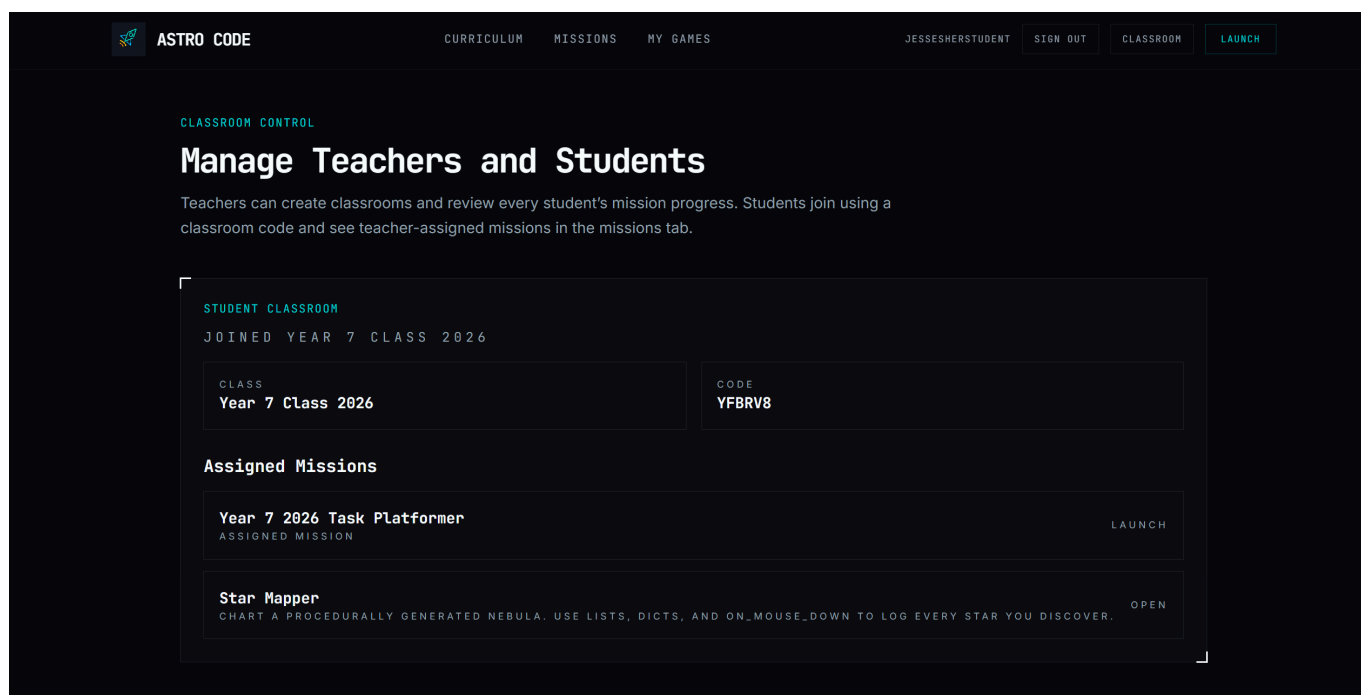
Purpose:

- To provide a place for the user to join a classroom via a code.

Key Elements:

- Navigation Bar
- Classroom Code Input box
- Classroom Join Button
- Footer

Classroom Page for Student After Joining



Purpose:

- To show the classroom they are in with its join code and their assigned tasks.

Key Elements:

- Navigation Bar
- Classroom Name
- Classroom Join Code
- List of Assigned Tasks
 - Open Button for Assigned Task
- Footer

6. Quality and Evaluation

6.1 Acceptance Criteria

- FR-1: User account creation completed
- FR-2: User account login completed
- FR-3: Teacher classroom creation completed
- FR-4: Student classroom joining via code completed
- FR-5: Teacher assignments task completed
- FR-6: Students can view assigned tasks completed
- FR-7: User can create a new game completed
- FR-8: User can upload assets for games completed
- FR-9: User can share game completed
- FR-10: Teacher can view status of students task completed

6.2 Testing Approach

Test Case	Expected Result	Actual Result	Pass
FR-1: User account creation	New user account created	New user account was created	✓
FR-2: User account login	User logs into account	User was able to log into their account	✓
FR-3: Teacher classroom creation	New classroom gets created	New classroom was successfully created.	✓
FR-4: Student classroom joining via code	Student will be able to join class	Student successfully joined the classroom	✓
FR-5: Teacher assignments task	The task will get assigned to the students	The task was successfully assigned to the students	✓
FR-6: Students can view assigned tasks	Students will be able to view their tasks	The tasks successfully appeared to the students.	✓

Test Case	Expected Result	Actual Result	Pass
FR-7: User can create a new game	User will be able to make a new game	The user successfully created a new game	✓
FR-8: User can upload assets for games	User will be able to upload custom assets to their game	The user's assets were successfully uploaded	✓
FR-9: User can share game	User's game will be shared via a link	The user's game was successfully shared and created a link	✓
FR-10: Teacher can view status of student's task	The teacher will be able to see the status of the student's task	The teacher successfully was able to view the student's task status	✓

6.3 Known Limitations and Future Work

Known Limitations

- 3D Games
- Different programming languages
- Live Code Edit Sharing

Future Improvements

- 3D Game functionality
- Able to programme in: Javascript, C++, C# etc.
- Live code sharing where teams are able to edit the same code live.

References

- GitHub Repository: <https://github.com/astrogames06/astro-code-app>